

Geometry Instructs Germ Layer Specification in Liver Organoids

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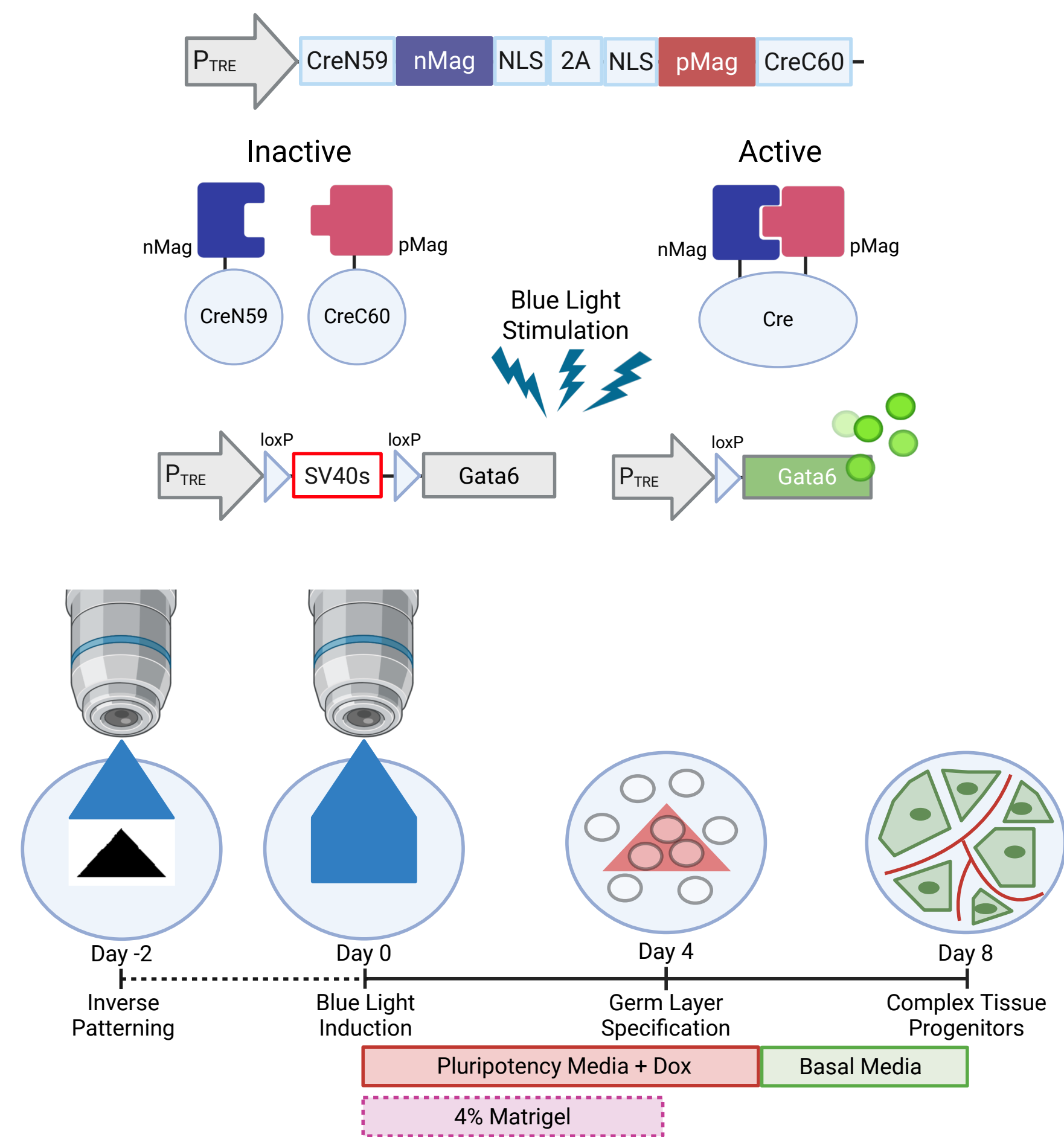
Background

- Variable transgenic **GATA6** induces fetal liver organoid self-organization in hiPSC¹.
- **Geometric constraints** improve reproducibility in organoid cultures².

Hypothesis

The **shape** and **size** of a developing liver organoid determines its tissue features.

Methods



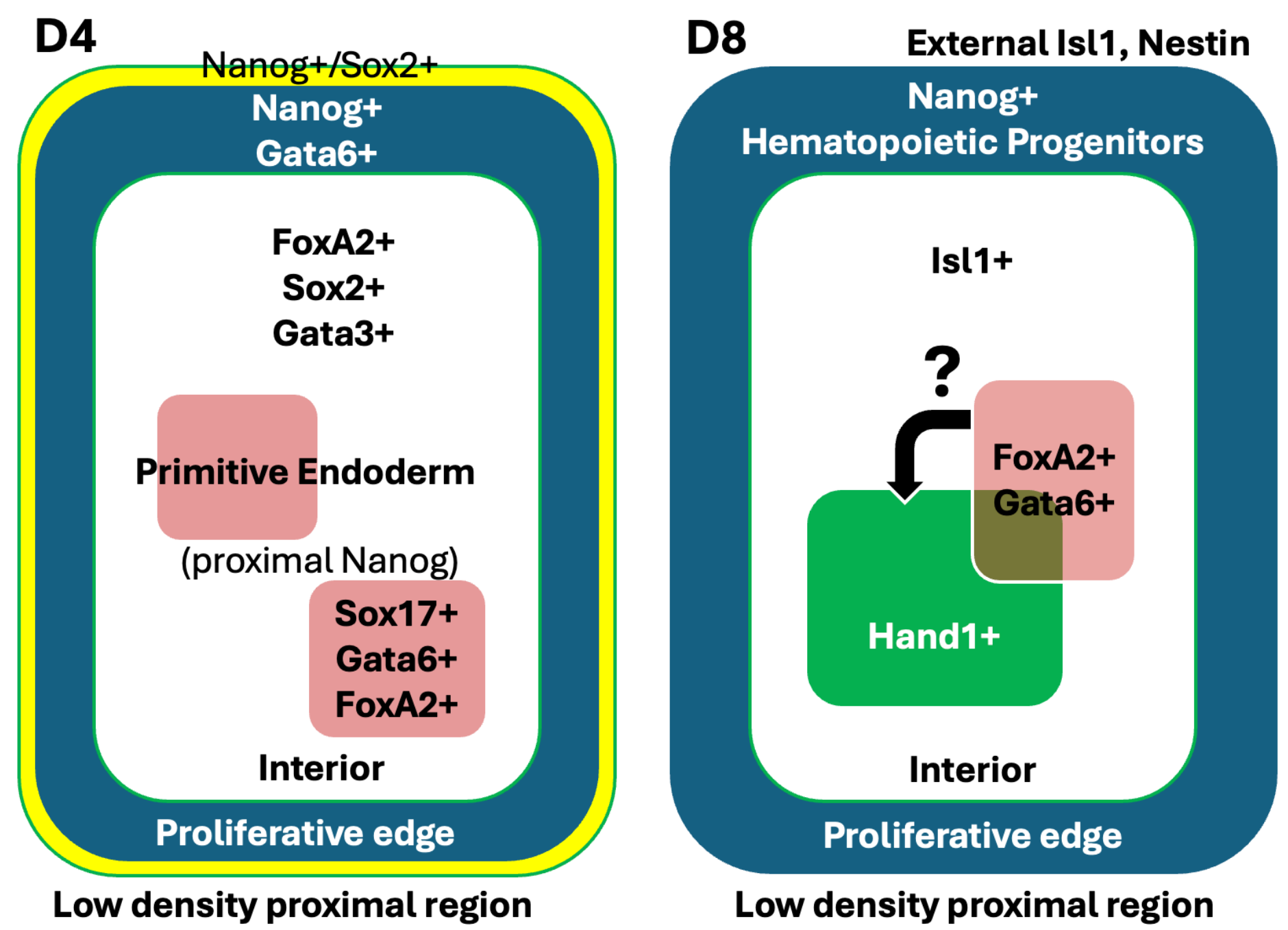
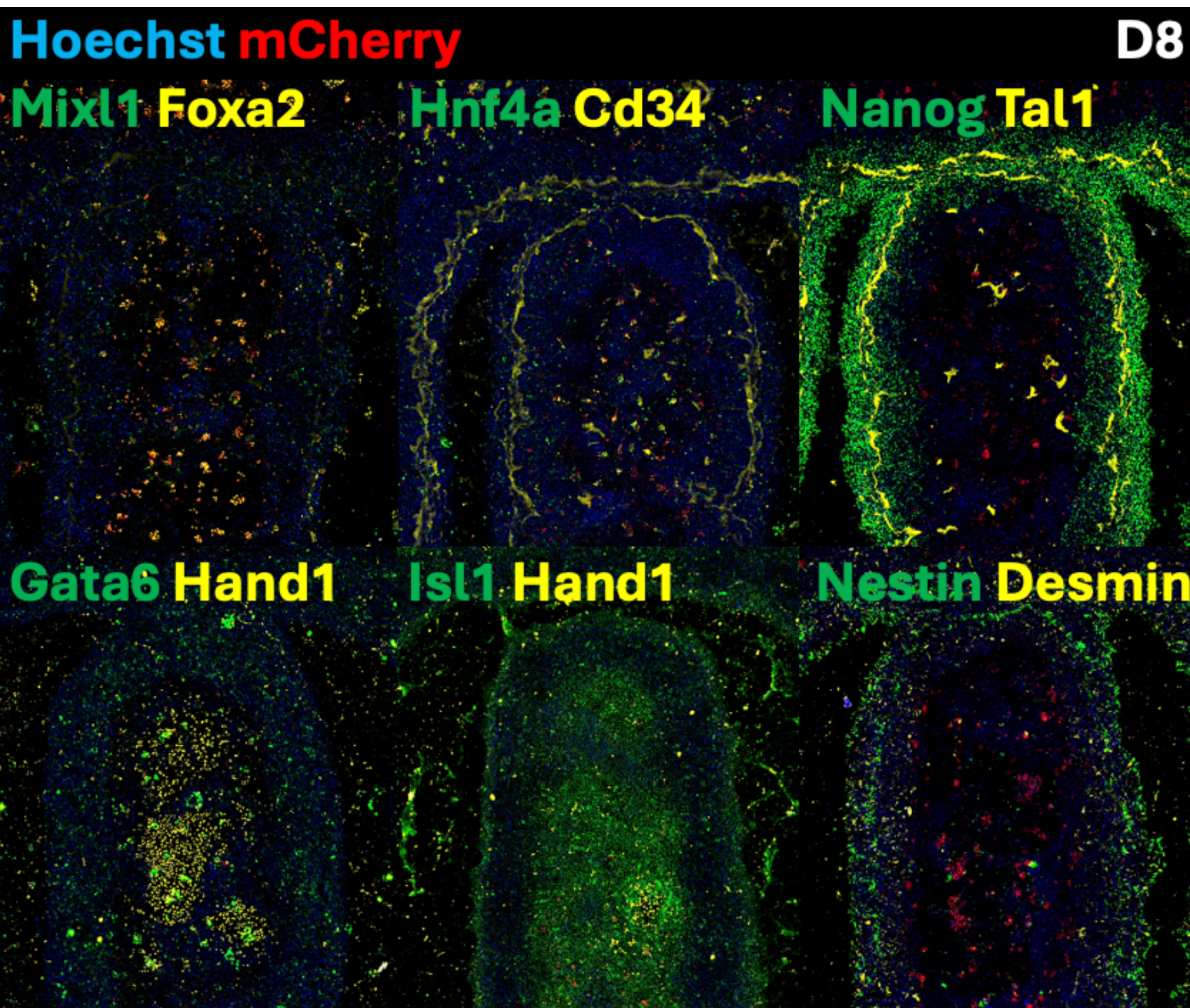
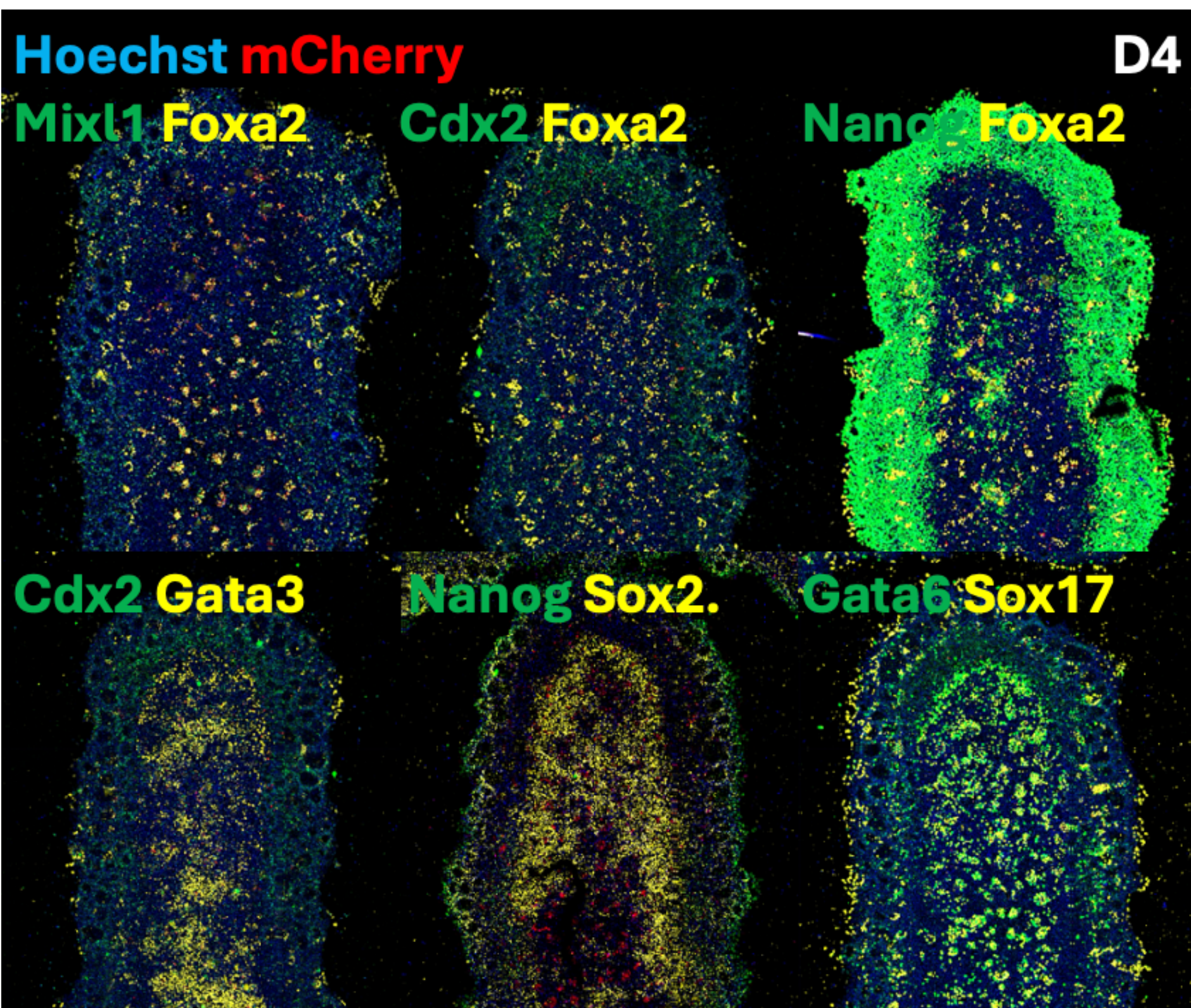
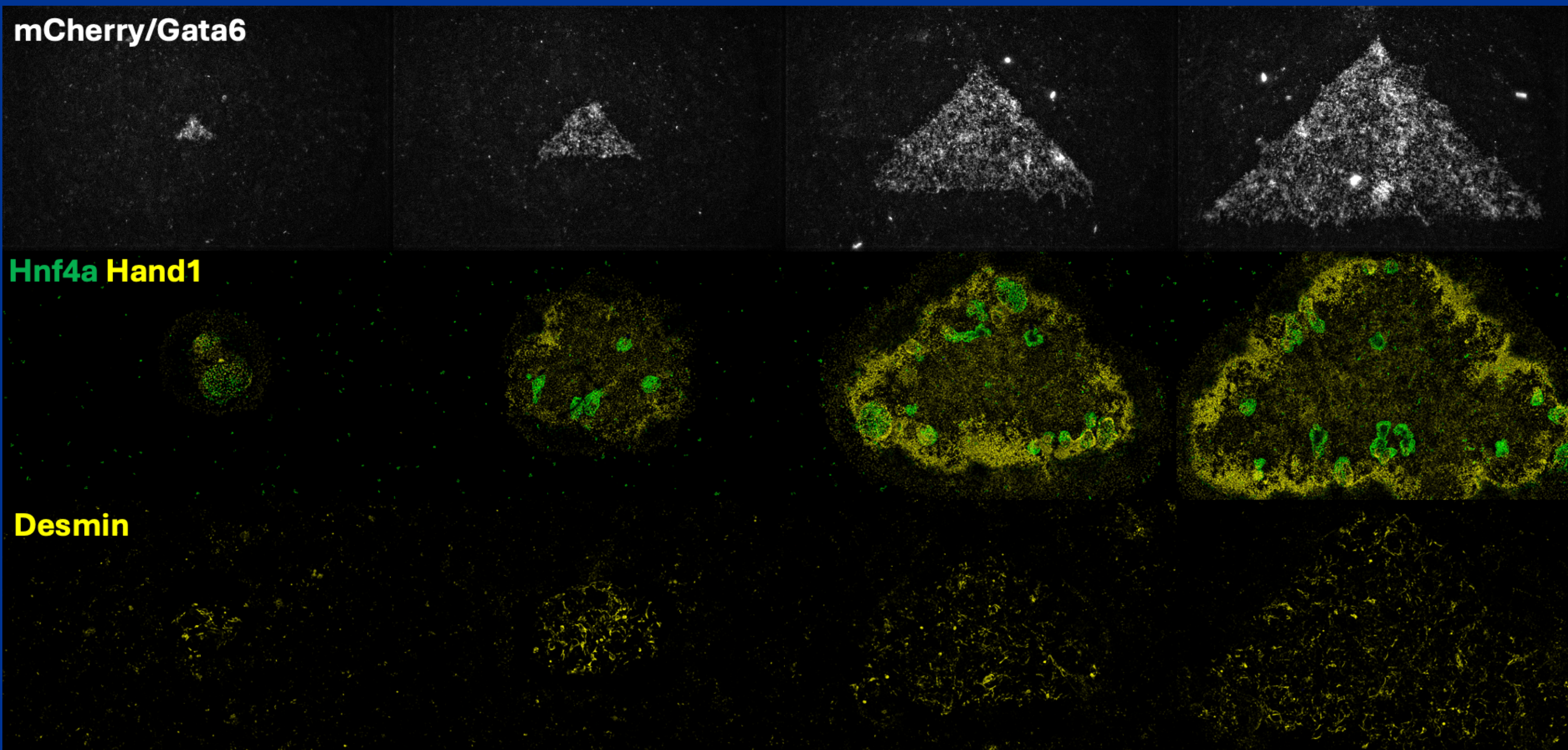
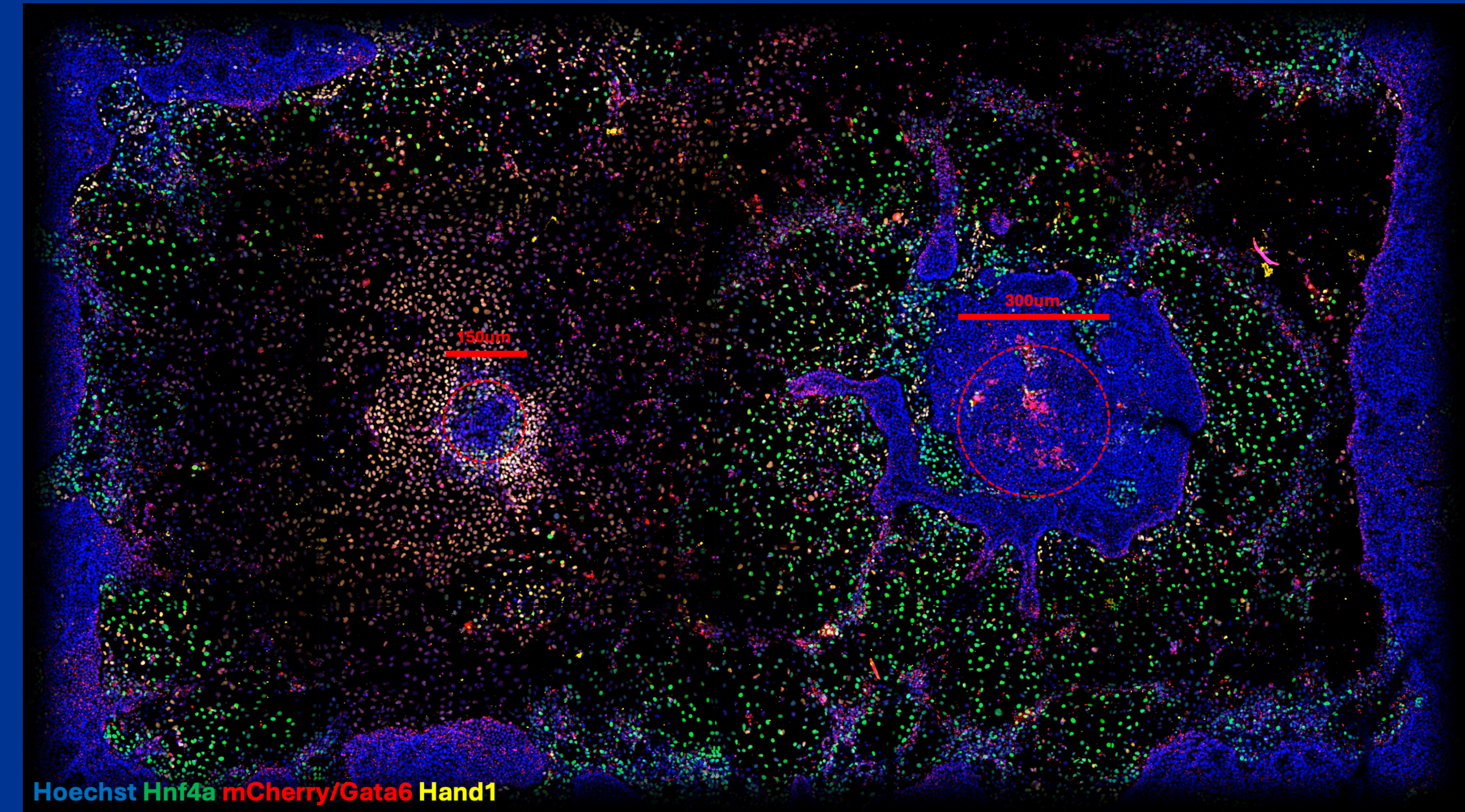
GATA6-expressing hiPSCs are induced with blue light on patterned substrates via our optogenetic microscopy platform.

Key Takeaways

- Patterns above a size threshold give rise to a proximal **Hand1+ boundary zone** and internal **Gata6+/Sox17+ islands**.
- 3D culture permits a highly proliferative proximal **Nanog+/Gata6+ zone**, which gives rise to **hematopoietic progenitors**.
- Pattern geometry may allow us to "paint" liver organoids by tissue type.



Developing liver organoids above a size threshold create their own extraembryonic tissue environment.



References and Acknowledgements

1. Guye, P., Ebrahimkhani, M., Kipniss, N. et al. Genetically engineering self-organization of human pluripotent stem cells into a liver bud-like tissue using Gata6. Nat Commun 7, 10243 (2016). <https://doi.org/10.1038/ncomms10243>
2. Gjorevski, N., Nikolaev, M., Brown, T. E., Lutolf, M. P. et al. Tissue geometry drives deterministic organoid patterning. Science, 375 (2022). <https://doi.org/10.1126/science.aaw9021>

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